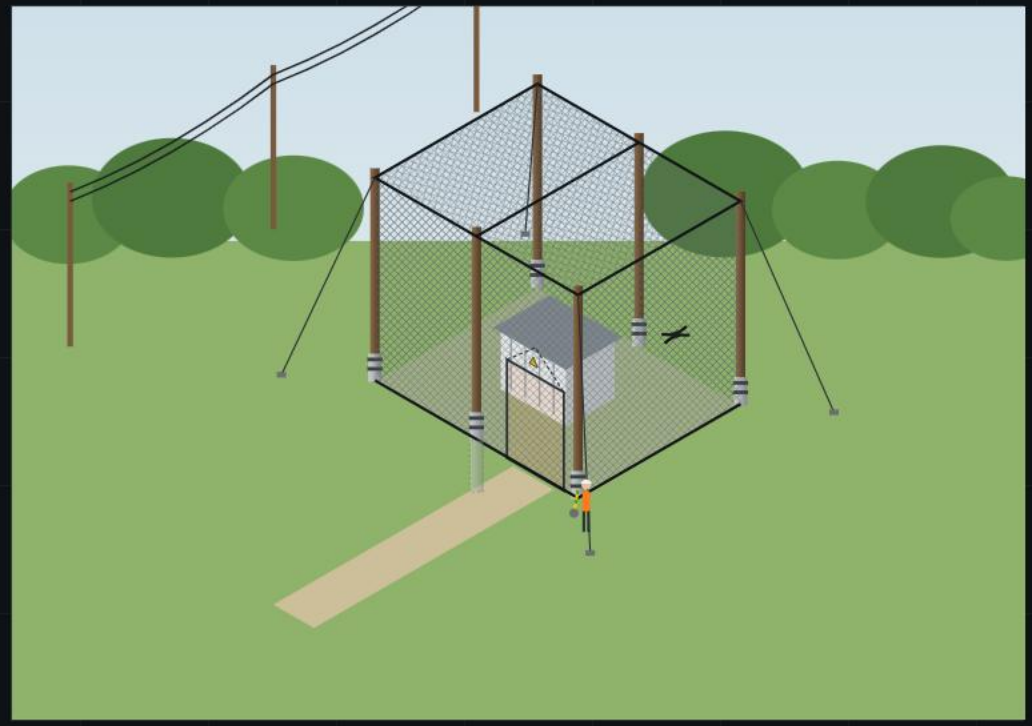
MATRICA

Matrica: drone nets from grid surplus.



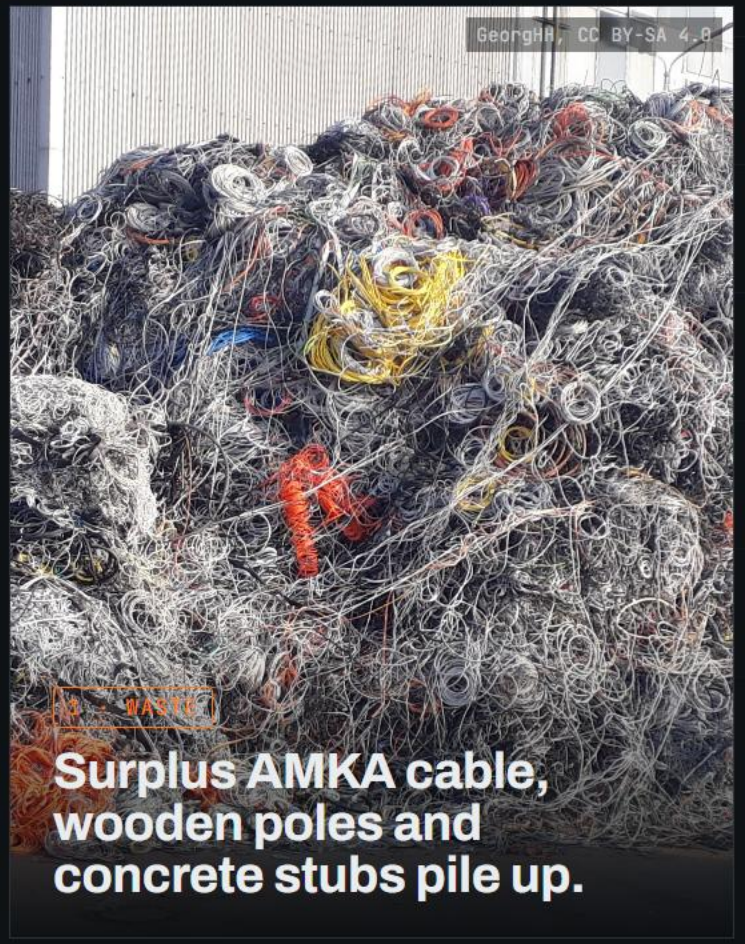
Tīkls no tīkla: in Latvian, *tīkls* means both **net** and **grid**.

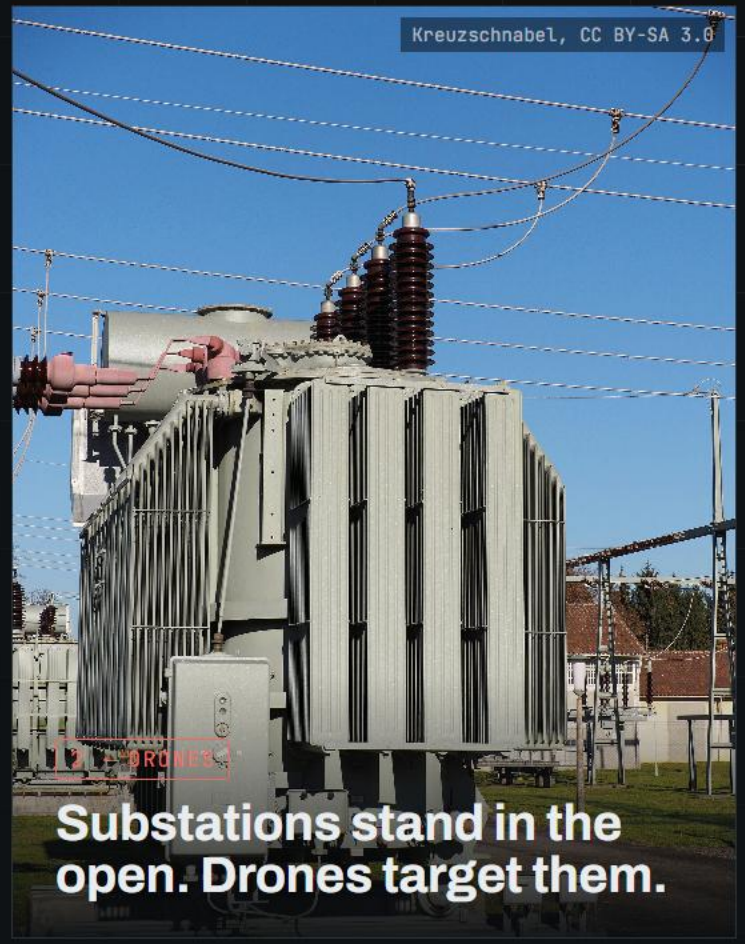
WE TAKE `AMKA cable` + `Wooden poles` + `Concrete stubs` → `Drone net`

`Climathon 2026 · RTU` `Challenge: AS "Sadales tīkls"`

THE PROBLEM

Three problems, one place.








BEFORE
Today: transformers stand unprotected.
Photo: Tiia Monto, CC BY-SA 4.0
MATRICA
Hackathon 2020 | Kaadetlas tikis challenge | 5/15



Photo: Tiia Monto, CC BY-SA 4.0 · cage visualisation: Matrica

AFTER · WITH MATRICA

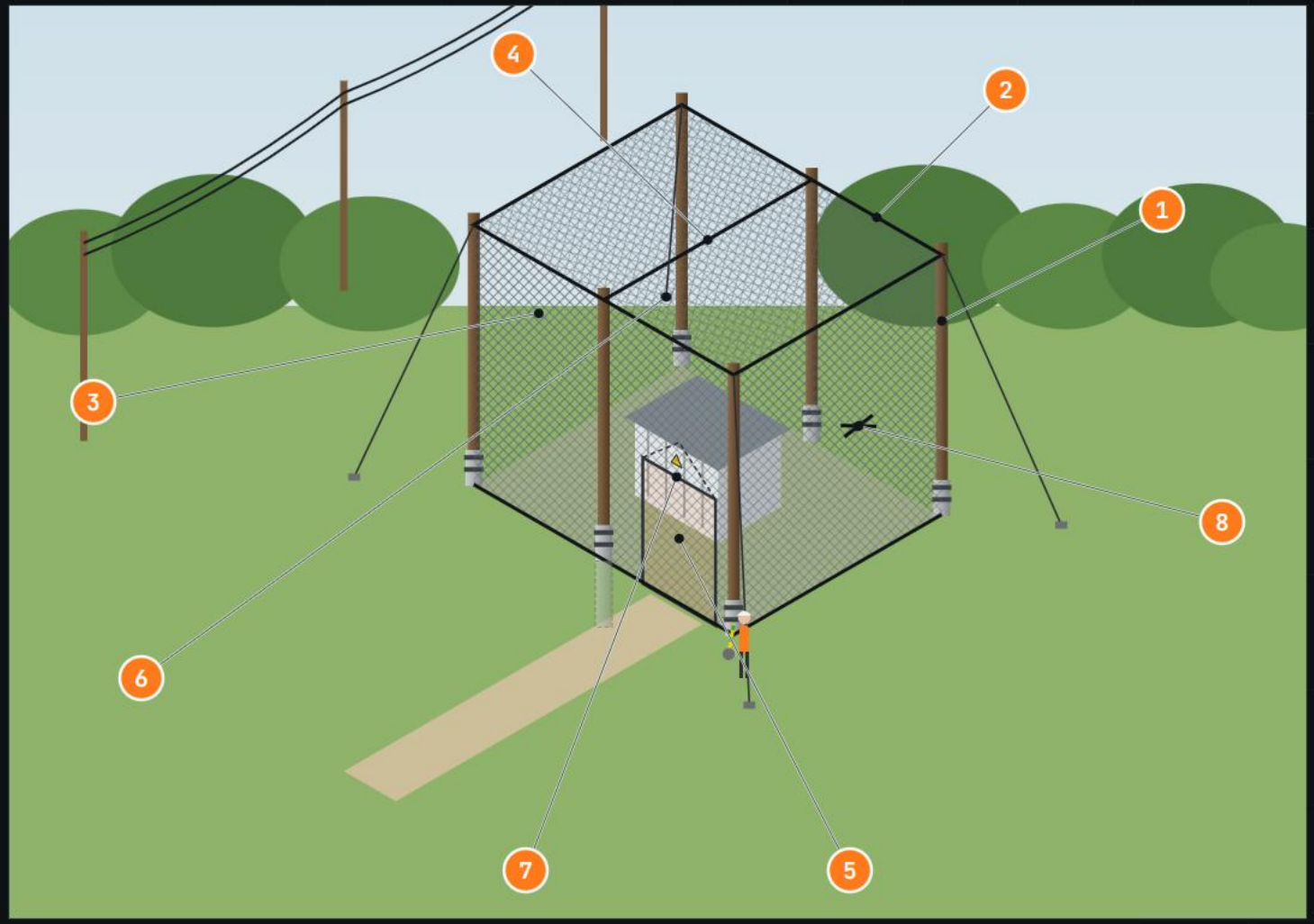
The whole substation inside a net from the grid.

Six masts from reused poles, woven AMKA net on every side and over the top. Same real photo; the cage is a visualisation.

MATRICA

THE PRODUCT

The structure: six masts, a woven net, a roof.

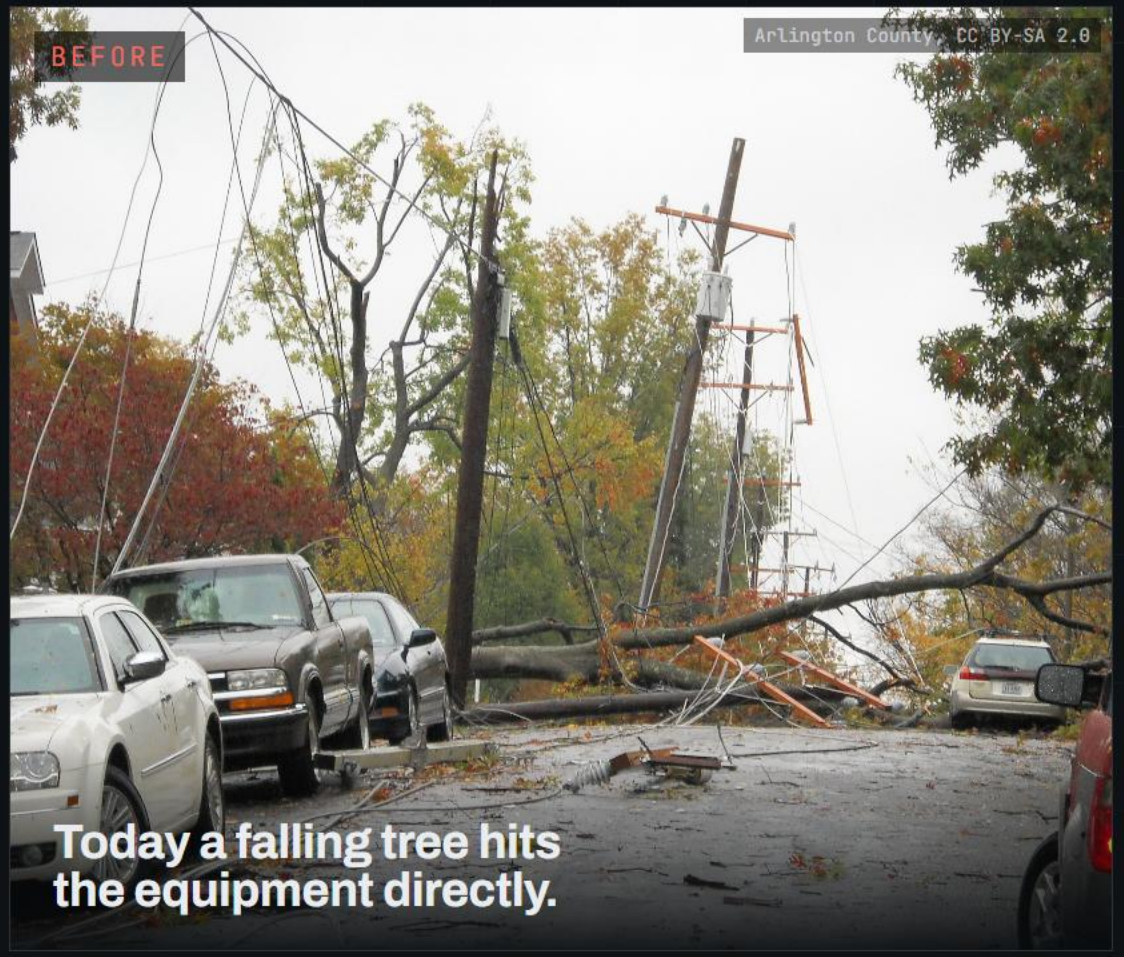


1. **Mast**
   Old wooden pole banded to a reclaimed concrete stub
2. **Top rope**
   AMKA cable, used whole, around the tops and across the roof
3. **Wall net**
   Woven from AMKA aluminium strands, 50 mm mesh, ~9 m high
4. **Roof net**
   Against drones diving from above
5. **Gate panel**
   Lifts for service vehicles
6. **Guy wire**
   To a ground anchor at each corner
7. **The protected substation**
   20/0.4 kV kiosk inside a 10 × 8 m enclosure
8. **A drone caught in the wall net**
   Metal mesh that propellers can't cut

20/0.4 kV kiosk · 10 × 8 m enclosure · net ~9 m high on all sides and over the top. The overhead line on the left is where the AMKA comes from.

STORMS

The net also takes the storm.



Guyed masts and the metal net catch branches before they reach the substation.

Checked after every storm and icing. Wet snow on the roof net is a known risk to test.

THE PRACTICAL PROCESS

The workshop makes the net. Crews raise it the way they already work.

IN THE WORKSHOP



01
Sort the AMKA
Keep long, undamaged lengths for ropes. Strip the rest.



02
Unwind the strands
Each phase core unwinds into aluminium strands of about 2–2.5 mm.

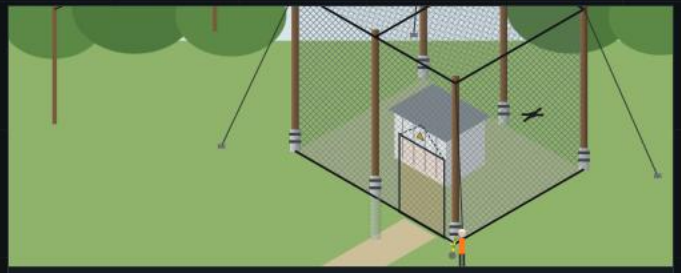


03
Weave the net
A chain-link machine turns the strands into 50 mm mesh panels.

ON SITE



04
Set the masts
Stub ~2.2 m into the ground, wooden pole banded on top. Grid crews do this every day.



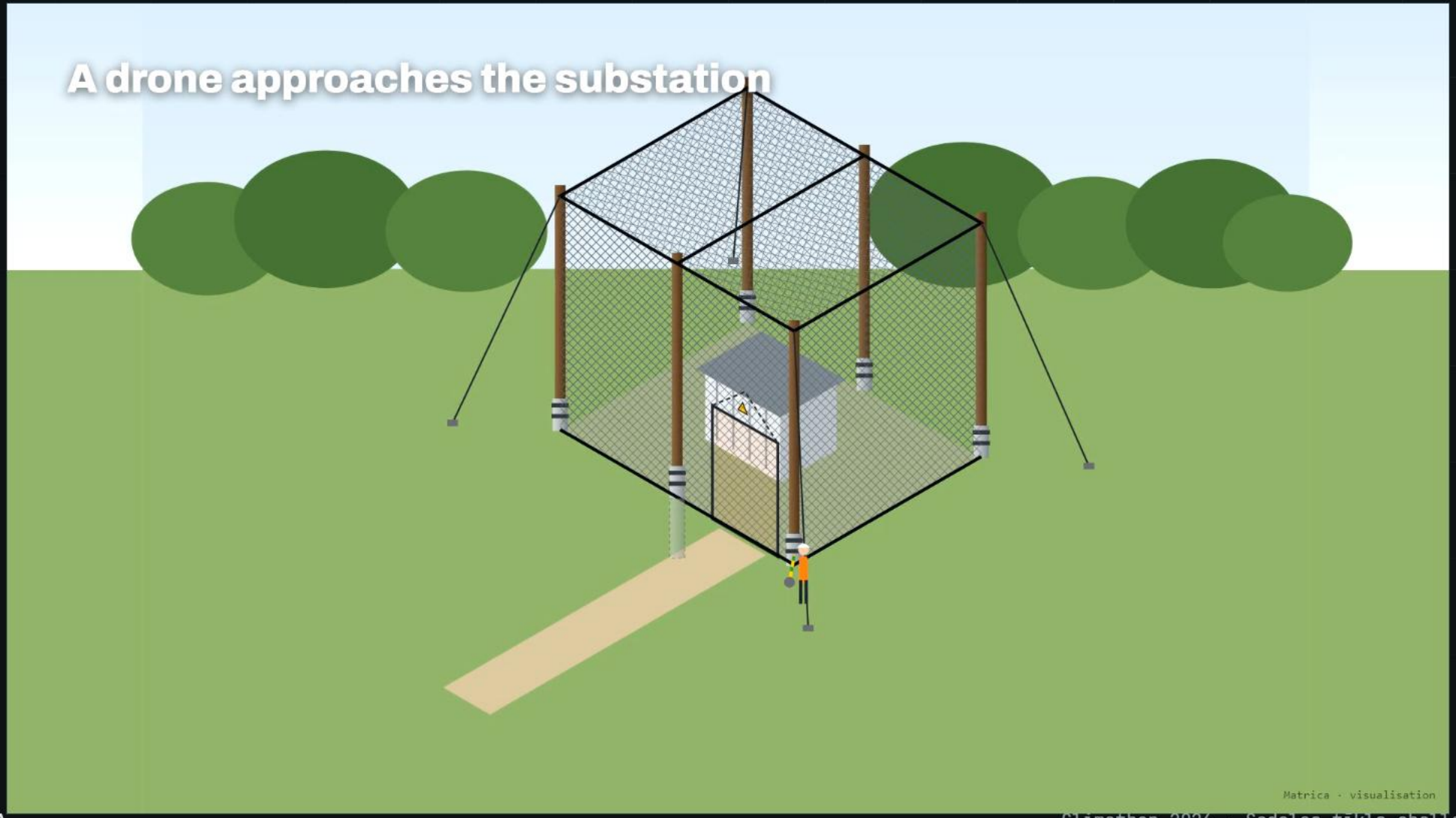
05
Hang ropes and net
AMKA around the tops and across the roof, guys at the corners, net laced on.



06
Earth and inspect
Bond the net to the site's earth. Check it after storms and icing.

Photos: GeorgHH (CC BY-SA 4.0); Tiia Monto (CC BY-SA 4.0) with AI visualisation. Drawings: Matrica, after the team summary.

The drone stops at the net, not at the substation.



WHERE

Where it can go after the pilot.

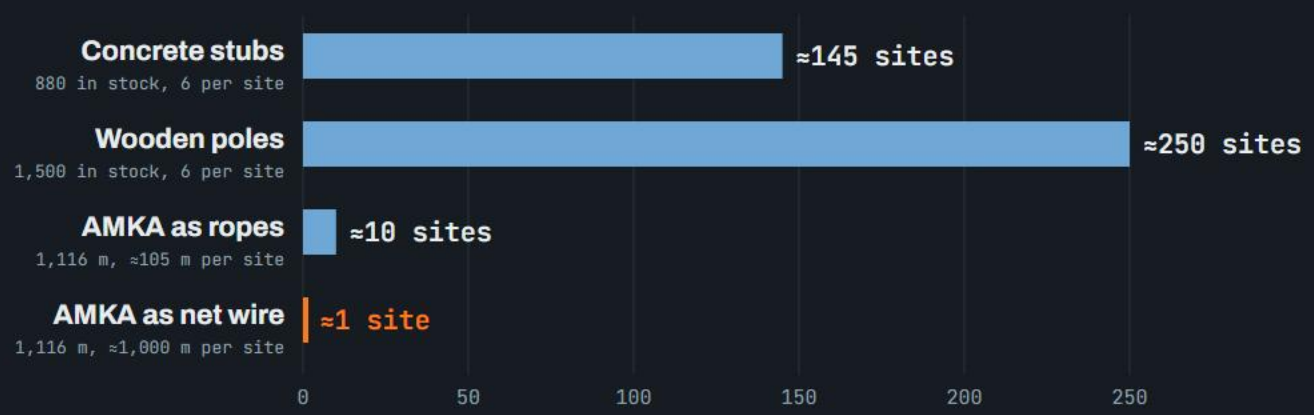


110 kV substation transformers	33
Grid battery storage	2
Hydro plant transformers	3
Heat and power plant transformers	2
Gas storage and stations	1
Broadcast transmitters	1
Airport navigation aids	2

The pilot is one 20/0.4 kV kiosk substation. These larger critical objects come later. Points at town level from public sources; exact sites to confirm with AST and Sadales tīkls.

THE NUMBERS · FROM THE SADALES TĪKLS STOCK SHEET

Poles for hundreds of sites. Cable for one.



CABLE IS THE BOTTLENECK

So we start with **one site**.

The grid already reused 487 stubs, 444 poles and 374 m of AMKA in 2026. We take only the leftover surplus.

ONE SITE NEEDS ESTIMATES

6	~105 m	~1 km	~390 m²	~20 km
masts	AMKA for ropes	AMKA for the net	net (36 × 8.7 m + 80 m² roof)	aluminium strand

DOES IT HOLD?

A 1 kg FPV drone at 30 m/s carries **~450 J**.
One 2.5 mm strand holds **~800 N**.

Test: a 10 kg sandbag dropped from 4.6 m gives the same 450 J on a panel.

BUDGET · ESTIMATES

€22–35k to prove it on one substation.

Initial capex · workshop — €12–19k

Semi-automatic chain-link machine $4–8k ex-factory + shipping, VAT, set-up for aluminium wire	€8–12k
Cable stripping machine + hand tools for AMKA cores	€1.5–3k
Strand unwinding and straightening rig reel stands, straightener, workbench	€0.5–1.5k
Test rig: panel frame + 10 kg sandbag drop 450 J impact test	€0.5–1k
Safety gear, earthing tester, tools	€1–2k

One-off. The same workshop serves every later site.

Pilot site · one kiosk substation — €10–16k

Reused stubs, poles, AMKA Sadales tīkls surplus (scrap value forgone ~€1k)	€0
Bought hardware steel bands, guys + 4 anchors, clamps, lacing wire, gate frame, gravel	€2.5–4k
Earthing set conductor, clamps, rod, measurement	€0.4–0.8k
Workshop labour, ~70–100 h sort, strip, unwind, weave ~390 m² at €15–20/h incl. tax	€1–2k
Site crew 3 × 4 days plus crane truck ~2 days (€39–45/h) and auger	€1.5–2k
Crane, auger, transport	€1–1.5k
Engineering and safety check structure, clearance, earthing, permit	€2–4k
Contingency 15%	€1.3–2.1k

TO START — Climathon prize €1,500 · Sadales tīkls pilot contract · State drone-protection pilot (KEM) · LIFE / dual-use grants

Prices: supplier listings, LV wage and crane-rental data, Oct 2026. To confirm with quotes.

CLIENTS · OUR PROPOSAL

Who uses it, and why they'd pay.

Pilot partner and first user

Owns the kiosk substations and the surplus. Gives material and a site; gets protection and reuse.

ST

Next users in Latvia

Own critical sites that drones and storms threaten: substations, grid batteries, plants, gas and broadcast.

AST

elektrum group

conexus BALTIC GRID

GASO

LVRTC

Next in the Baltics

Same equipment, same threat, same kind of surplus.

eso

Litgrid

Elektrilevi

elering

WHY THEY PAY

Material from their own surplus, crews that already know the work, a metal net propellers can't cut, and a resilience measure the EU CER Directive asks critical entities to take.

WHY NOW · 22 SEP 2026

The Cabinet approved a pilot with energy companies to protect energy infrastructure from drones: shared requirements, joint procurement, Latvian solutions welcome. AST gets €18.5M in EU funding for perimeter and counter-drone protection.

The team summary names no client; this is our proposal. No agreements yet. Logos belong to their owners. "Why now": LV portāls, KEM, 2026.

PRACTICAL

A concept, tested before it goes near live equipment.

Safe

Net bonded to earth

Clear of live parts

Far enough that a blast in the net can't reach equipment

Honest limits

Stops FPV and small multirotors, not large one-way attack drones

Wet snow can bridge the mesh and load the roof

AMKA strands may crack in the weaving machine

Pilot

One 20/0.4 kV kiosk site

Weave test samples, then a sandbag test

An agreed limit on how much surplus we take

TARGET SUPPLIERS, CUSTOMERS AND PARTNERS

The companies that would want this.

MATERIAL SUPPLIERS AND BUYERS

ST	AST	elektrum group	conexus BALTIC GRID	GASO	LVRTC
Tet	LMT	Tele2	Bite	LDz	Elektrilevi
elering	Telia	eso	Litgrid	ignitis grupė	

DELIVERY PARTNERS

JAUDA	ELTEL	VINCI ENERGIES	Eco Baltia	NKT	prysmian	SAF Tehnika	Origin Robotics

EUROPEAN FUNDS FOR SCALE-UP

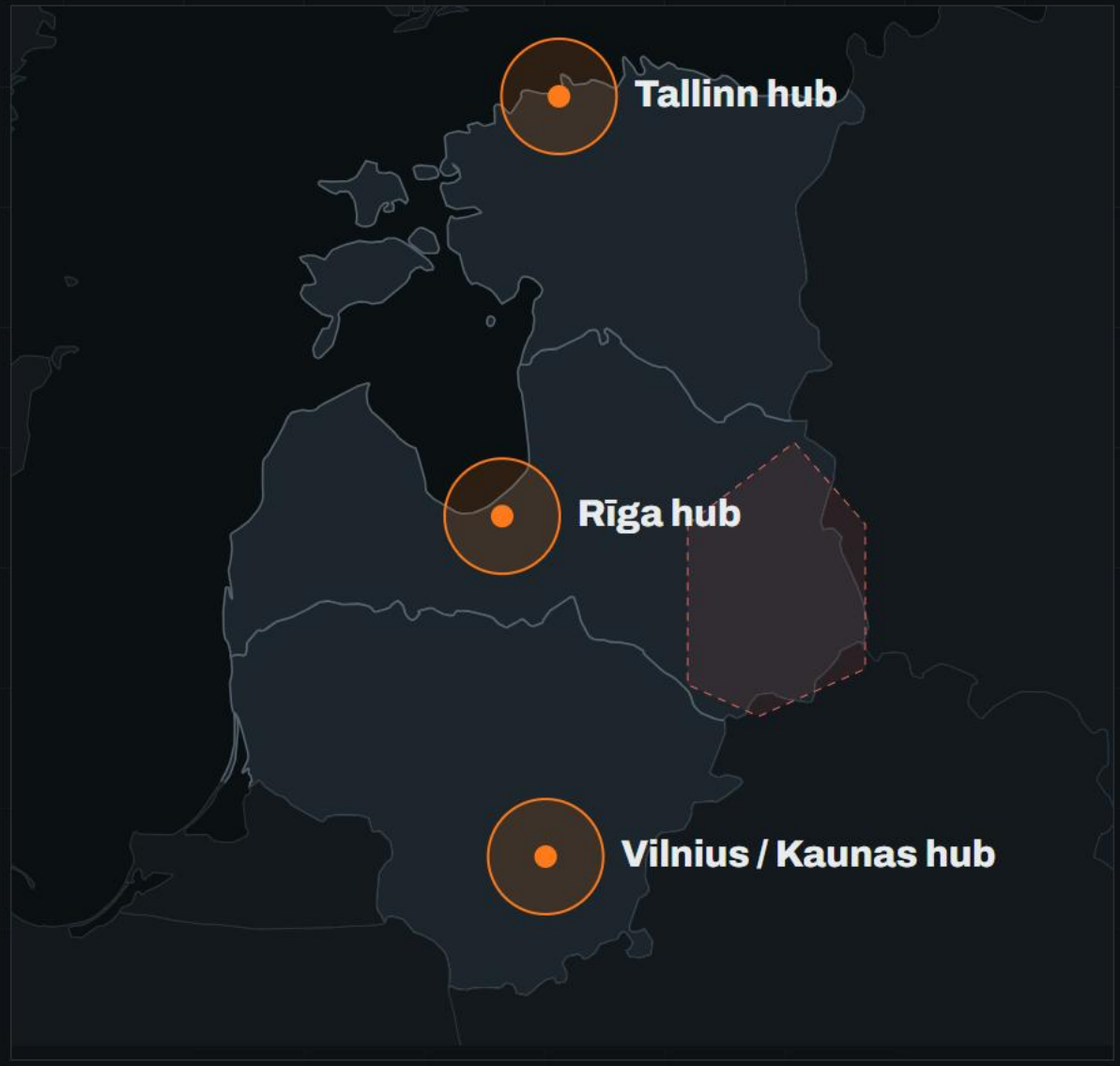
- **Latvian dual-use programme** LIAA / ALTUM, EU co-financed
- **LIFE** Circular economy, material reuse
- **Horizon Europe Cluster 3** Critical infrastructure protection
- **EIC Accelerator** Grant + equity to scale
- **Interreg Baltic Sea Region** Three-country rollout

No agreements yet. Logos belong to their owners and are shown to name target companies.

THE FUTURE

A Baltic take-back network.

Once the pilot works: three hubs, every grid company's surplus as material, each country's surplus protecting its own grid.



OUR ASK

A one-site pilot with Sadales tīkls, and an agreed share of the surplus.

APPENDIX

Scored target objects (0–10 each)

Object	Owner	Crit	Fit	Exp	Vol	Safe	Pay	/60
110/20 kV substation transformers	AST / Sadales tīkls	9	8	7	8	5	9	46
20/0.4 kV kiosks feeding hospitals and water	Sadales tīkls	6	10	5	9	8	7	45
Grid battery storage containers	AST	9	9	8	2	7	9	44
Gas regulating and metering stations	Conexus / Gaso	7	9	6	7	8	7	44
Drinking-water pumping stations	Municipal utilities	8	9	5	8	9	5	44
Mobile base station cabinets	LMT / Tet / Tele2 / Bite	7	8	6	10	8	5	44
Broadcast transmitter sites	LVRTC	8	7	6	4	7	7	39
Railway traction substations	LDz	7	8	6	5	6	6	38
Small district-heating boiler houses	Municipal	7	6	5	7	8	5	38
Data-centre cooling and generator yards	Tet and others	7	8	4	4	8	7	38
Synchronous condensers	AST	9	5	6	2	6	9	37
Airport navigation aids	LGS	8	7	4	3	8	7	37

Assumptions behind the numbers

- Only the stock figures come from the Sadales tīkls sheet: 880 stubs, 1,500 poles, 1,116 m AMKA (reused in 2026: 487, 444, 374 m).
- One site: 20/0.4 kV kiosk, 10 × 8 m enclosure, 6 masts, ~105 m AMKA ropes, ~1,000 m AMKA for the net, ~390 m² of net, ~20 km of strand (~50 m per m²).
- Impact: 1 kg FPV at 30 m/s ≈ 450 J; one 2.5 mm strand ≈ 800 N; test with a 10 kg sandbag from 4.6 m.
- A hackathon concept, not a design: an engineer and tests must verify it before anything is built near live equipment.

Sources

- Sadales tīkls challenge brief and stock list
  `Climathon 2026 RTU materials (this repo)`
- Ukraine: three levels of energy protection
  `https://war.ukraine.ua/war-news/ukraine-protection-russias-attacks-energy-infrastructure/`
- EIB €86M to Ukrenergo for anti-drone protection
  `https://en.interfax.com.ua/news/general/1036088.html`
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- Lithuania's Gižai substation, €48M+ hardening
  `https://eutoday.net/lithuania-drone-proof-harmony-link-eu-grid-security/`
- AST batteries in Rēzekne and Tume
  `https://www.ess-news.com/2025/10/30/latvia-adds-big-batteries-to-complete-grid-sync-with-europe-two-major-estonian-projects-secure-finance`
- Anti-drone net mesh sizes and twine ratings
  `https://ahcebrian.com/en/blog/passive-anti-drone-nets-explained/`
- EU CER Directive, Latvian transposition
  `https://www.iem.gov.lv/en/article/government-strengthens-critical-infrastructure-resilience-and-national-security`
- Baltic defence and dual-use investors
  `https://www.vestbee.com/insights/articles/europes-defence-and-dual-use-investors`
- KEM: unified approach to energy critical infrastructure security (Cabinet pilot 22 Sep 2026; AST €18.5M EU funding)
  `https://lvportals.lv/dienaskartiba/394848-kem-tiks-ieviesta-vienota-pieeja-energetikas-kritiskas-infrastrukturas-drosibai-2026`
- Full research with all sources
  `research/grid-shield.html`

Image and data credits

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- **Substation before/after:** Tiia Monto, CC BY-SA 4.0, Wikimedia Commons: Milovice-transformer station.jpg. The after image is an AI edit with Qwen-Image-Edit, shared under CC BY-SA 4.0.
- **110/20 kV transformer:** Kreuzschnabel, CC BY-SA 3.0, Wikimedia Commons: Transformer 110 kV 20 kV.jpg
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- **Hydraulic cable cutter:** S.J. de Waard, CC BY-SA 3.0, Wikimedia Commons: Cable cutting device.JPG
- **Country outlines:** mledoze/countries (GitHub), ODbL, from Natural Earth
- **Company logos:** Wikimedia Commons and the companies' own websites. Logos belong to their owners.
- **Explainer video:** Matrica, made with Remotion (github.com/remotion-dev/remotion)